

Workshop 1

Requirements Analysis and User Stories

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1. Requirements

Main functional (FR) and non-functional (NFR) requirements of VetCare, a centralized web platform for veterinary clinics. Requirements are grouped by module; quality attributes that apply to the whole system are listed at the end.

Module 1: Authentication & Access Control (RBAC)

- FR_01** When a new Pet Owner submits the registration form, or activates an invitation issued by the clinic, the system must create an account and store their credentials to grant authorized access to the owner portal.
- FR_02** When a registered user inputs their credentials on the login page, the system must authenticate the credentials and establish an active session to allow entry into the platform.
- FR_03** When an authenticated user accesses any page or endpoint, the system must evaluate their assigned role (Veterinarian, Receptionist, Administrator, Veterinary Assistant, or Owner) to restrict execution to role-permitted actions only.
- FR_04** When a user requests a password reset with their registered email, the system must send a time-limited reset link so that the user can set a new password.
- FR_05** When an authenticated user logs out, or when a session expires due to inactivity, the system must invalidate the session and require a new login.
- FR_06** When an Administrator creates, edits, or deactivates a staff account and assigns a role, the system must save the change and apply the new permissions from the user's next request.
- NFR_01** *Security.* When a user authenticates or sends a request, the system must store passwords as salted hashes (e.g., bcrypt or Argon2), transmit all data over TLS, and validate signed JWT tokens with an expiration time to prevent credential exposure and session hijacking.
- NFR_02** *Security & Compliance.* When any API call is received, the system must verify role permissions at the backend layer to guarantee data confidentiality across all clinical and administrative records.

Module 2: Patient & Client Management (CRM & Clinical Records)

- FR_07** When a Receptionist enters a new customer's personal details and preferred communication channel, the system must save the owner profile to establish a master client record in the database and offer to send an invitation to activate the owner's portal account.
- FR_08** When a Receptionist associates a new pet with an existing owner ID, the system must link the pet profile to the owner record to organize multi-pet client accounts.

- FR_09** When a Veterinarian opens a new consultation file for a patient, the system must load species-specific medical fields (for dogs, cats, or exotic animals) to capture clinical parameters relevant to that species.
- FR_10** When a Veterinarian or Veterinary Assistant views a patient's full medical file, the system must display complete clinical records (vital signs, diagnoses, treatments, dosages, evolution notes, and surgical history) to support clinical decision-making.
- FR_11** When a Pet Owner accesses their pet's health portal, the system must render a simplified medical summary (basic info, vaccination status, allergies, and active treatments) to inform the owner without exposing technical veterinary jargon or internal notes.
- FR_12** When an authorized user opens a patient's medical background, the system must retrieve the complete vaccination history, recorded allergies, and previous consultations to review past care.
- FR_13** When an authorized staff member searches by owner name, national ID, or pet name, the system must return the matching owner and patient records so that they can be located quickly.
- FR_14** When a Veterinarian or Veterinary Assistant records a vaccine application (vaccine, lot number, and date), the system must store it in the patient's vaccination history and calculate the next due date.
- FR_15** When a Veterinarian issues a prescription with drug, dose, frequency, and duration, the system must save it in the consultation record and link it to the corresponding inventory item.

Module 3: Space & Appointment Management

- FR_16** When a Pet Owner submits an online booking request with a selected pet, service type, and time slot, the system must log the appointment request in pending status and notify the reception team to initiate the scheduling workflow.
- FR_17** When a Receptionist confirms, reschedules, or cancels an appointment request, the system must update the schedule state and reflect real-time availability to prevent scheduling conflicts.
- FR_18** When a Receptionist assigns a consultation room to an appointment, the system must check doctor and room availability to prevent double-booking of physical spaces or veterinary personnel.
- FR_19** When an Administrator defines a veterinarian's weekly working hours and time off, the system must save the schedule and use it to calculate the available appointment slots.
- FR_20** When a Pet Owner requests to cancel or reschedule a pending or confirmed appointment, the system must update its status, release the previously held slot, and notify the reception team.

- FR_21** When a user views the hospitalization module, the system must render a visual occupation map with status indicators (Critical, Stable, Under Observation) to display current bed and cage availability in real time.
- FR_22** When a Veterinarian admits a patient to hospitalization, the system must assign an available cage and set an initial status; when the patient is discharged, the system must release the cage and store the discharge date and summary.
- FR_23** When a Veterinarian or Veterinary Assistant performs a hospitalization round, the system must record the staff member's ID, timestamp, and observations to maintain an accurate patient evolution log.
- FR_24** When a Veterinarian changes the status of a hospitalized patient, the system must update the status indicator on the occupation map and log the change with its author and timestamp.
- FR_25** When an appointment is confirmed or an upcoming date approaches, the system must dispatch an automated confirmation or reminder message via the owner's preferred channel to reduce no-show rates.
- FR_26** When a pet's next vaccine due date approaches, the system must send a reminder to the owner via their preferred channel so that vaccination schedules are kept up to date.
- NFR_03** *Performance.* When a user interacts with the visual occupation map or appointment calendar, the system must update and render changes within 2 seconds to ensure seamless real-time operation.
- NFR_04** *Availability.* When users access the application during regular operational hours, the system must remain operational with 99.5% uptime to support continuous clinical workflow.

Module 4: Pharmacy & Inventory Management

- FR_27** When a user inspects pharmaceutical stock, the system must display available quantities categorized by batch/lot number and expiration date to ensure proper inventory tracking.
- FR_28** When an Administrator registers a purchase receipt (items, quantities, lot numbers, expiration dates, and supplier), the system must add the received items to the inventory.
- FR_29** When a Veterinarian prescribes a drug or records a vaccine application, or a Veterinary Assistant registers consumed supplies, the system must automatically deduct the corresponding quantities from the inventory to keep stock levels synchronized.
- FR_30** When a medication or vaccine drops below its minimum threshold or reaches its expiration period, the system must issue automatic alerts to administrative staff to prompt timely restocking.

- FR_31** When an Administrator creates or edits an inventory item or a supplier (including the item's minimum threshold and linked supplier), the system must save the record so that alerts and reordering use up-to-date data.
- FR_32** When an Administrator reviews a low-stock medication entry, the system must present the supplier contact details linked to that item to streamline reordering procedures.
- NFR_05** *Data Integrity.* When concurrent inventory deductions occur, the system must execute atomic database transactions to maintain accurate stock records without race conditions.

Cross-cutting Non-Functional Requirements

- NFR_06** *Usability & Design.* When a user accesses the web interface on any device, the system must adapt its layout dynamically, remaining fully usable on the latest versions of Chrome, Firefox, Edge, and Safari.
- NFR_07** *Localization.* When a user opens the application, the system must present the interface and notifications in Spanish by default.
- NFR_08** *Auditability & Traceability.* When a user views or modifies a clinical record, the system must log the user, action, and timestamp; clinical entries must never be deleted, and any correction must be stored as a new version that preserves the previous one.
- NFR_09** *Scalability.* When the number of users grows, the system must support at least 100 concurrent users without degrading the response times defined in NFR_03 and allow scaling its components independently.
- NFR_10** *Maintainability.* When code is pushed or a pull request is opened, the CI/CD pipeline (GitHub Actions) must build the project and run the automated tests, so that changes can be integrated and deployed safely.

2. User Stories

User stories and acceptance criteria for all relevant roles, written in the format "As a [role], I want to [action] so that [benefit]" with criteria in Given / When / Then form.

Pet Owner / Client

US_OWNER_01: Online Appointment Request

User Story: As a Pet Owner, I want to request an appointment online by selecting my pet, service type, and preferred slot so that I can schedule care without making a phone call.

Acceptance Criteria:

- Given** the owner is logged in and on the appointment scheduling page,
- When** they select their pet, the service type, and an available time slot, and submit the request,
- Then** the system registers the appointment as "Pending Confirmation" and notifies the reception team.

US_OWNER_02: View Simplified Medical Sheet

User Story: As a Pet Owner, I want to view my pet's simplified medical profile and vaccination record so that I can track my pet's health and upcoming vaccinations without confusing technical terms.

Acceptance Criteria:

- Given** the owner is logged into their dashboard,
- When** they navigate to their pet's profile,
- Then** the system displays basic pet details, upcoming vaccine dates, allergies, and last visit summaries, while excluding internal veterinary evolution notes.

US_OWNER_03: Create My Portal Account

User Story: As a Pet Owner, I want to create my account (or activate the invitation sent by the clinic) so that I can see my pets' information and book appointments online.

Acceptance Criteria:

- Given** the owner is on the registration page or has opened an activation link,
- When** they submit a valid email, a password, and their personal details,
- Then** the system creates the account with the Owner role and grants access to the owner portal.

US_OWNER_04: Cancel or Reschedule My Appointment

User Story: As a Pet Owner, I want to cancel or reschedule my appointment online so that the clinic can free the slot and I avoid making a phone call.

Acceptance Criteria:

- Given** the owner has a pending or confirmed appointment,
- When** they choose to cancel or reschedule it and confirm the action,
- Then** the system updates the appointment status, releases the slot, and notifies the reception team.

US_OWNER_05: Receive Appointment and Vaccine Reminders

User Story: As a Pet Owner, I want to receive reminders through my preferred channel so that I do not miss appointments or vaccine doses.

Acceptance Criteria:

- Given** the owner has a confirmed appointment or a vaccine due date approaching,
- When** the reminder date is reached,
- Then** the system sends a message via the owner's preferred channel with the pet name and the appointment date and time or the vaccine name.

Receptionist

US_RECEPTIONIST_01: Owner and Pet Registration

User Story: As a Receptionist, I want to register a new owner and link their pet(s) to their account so that the clinic has accurate contact details and patient associations.

Acceptance Criteria:

- Given** the receptionist is on the registration screen,
- When** they enter the owner's contact information, preferred communication channel, and pet details,
- Then** the system creates the owner profile and links the pet(s) under a single customer record.

US_RECEPTIONIST_02: Manage Appointments & Room Assignment

User Story: As a Receptionist, I want to confirm or reschedule pending appointments and assign an available consultation room so that there are no room or schedule conflicts for veterinarians.

Acceptance Criteria:

- Given** a pending appointment request exists,
- When** the receptionist approves it and assigns an available consultation room and veterinarian,
- Then** the system updates the appointment status to "Confirmed" and ensures no double-booking occurs for that room or doctor.

US_RECEPTIONIST_03: Search Owners and Patients

User Story: As a Receptionist, I want to search owners and pets by name or ID so that I can find their records quickly.

Acceptance Criteria:

- Given** the receptionist is on the client search page,
- When** they enter an owner name, national ID, or pet name,
- Then** the system lists the matching owners with their linked pets and lets the receptionist open the selected record.

Veterinarian

US_VET_01: Species-Specific Medical Record Entry

User Story: As a Veterinarian, I want to create and update a medical record tailored to the patient's species so that I can document vital signs, diagnoses, and treatments accurately.

Acceptance Criteria:

Given a veterinarian is conducting a consultation with a specific animal (e.g., cat or exotic animal),

When they open the medical record form,

Then the system renders input fields customized for that species (e.g., species-relevant temperature ranges and vital sign parameters) and saves the entered clinical diagnosis and prescription.

US_VET_02: Hospitalization Monitoring & Rounds

User Story: As a Veterinarian, I want to view the hospitalization map and record round observations so that I can monitor critically ill patients in real time.

Acceptance Criteria:

Given the veterinarian is reviewing hospitalized patients,

When they select a cage on the occupancy map and submit round notes,

Then the system records the timestamped observation note and updates the patient status indicator (e.g., Critical, Stable, Under Observation).

US_VET_03: Vaccinations Record

User Story: As a Veterinarian, I want to register the vaccines applied to a patient so that the vaccination history and the next due dates are always up to date.

Acceptance Criteria:

Given a veterinarian has an open consultation for a patient,

When they record a vaccine with its name, lot number, and application date,

Then the system saves it in the patient's vaccination history, calculates the next due date, and deducts the dose from the inventory.

US_VET_04: Issue a Prescription

User Story: As a Veterinarian, I want to prescribe medication with dose, frequency, and duration so that the treatment is documented and the pharmacy stock is deducted automatically.

Acceptance Criteria:

Given a consultation is open,

When the veterinarian selects a drug from the inventory and enters dose, frequency, and duration,

Then the system saves the prescription in the medical record and deducts the dispensed quantity from stock.

US_VET_05: Admit and Discharge Hospitalized Patients

User Story: As a Veterinarian, I want to admit a patient to an available cage and discharge it when ready so that the occupancy map always reflects reality.

Acceptance Criteria:

Given there is at least one free cage,

When the veterinarian admits a patient selecting a cage and an initial status,

Then the system marks the cage as occupied and shows the patient on the occupancy map.

Given a patient is hospitalized,

When the veterinarian discharges the patient,

Then the system releases the cage and stores the discharge date and summary.

Veterinary Assistant

US_ASSISTANT_01: Record Daily Care & Supplies Consumption

User Story: As a Veterinary Assistant, I want to log daily monitoring notes and supplies consumed during patient care so that patient status is kept up to date and pharmacy stock is accurately maintained.

Acceptance Criteria:

Given an assistant completes a routine check or routine care procedure,

When they enter daily monitoring values and mark consumed supplies (e.g., syringes, bandages),

Then the system saves the monitoring sheet update and automatically deducts the consumed items from inventory.

Clinic Administrator

US_ADMIN_01: Pharmacy Stock Alerts & Supplier Lookup

User Story: As a Clinic Administrator, I want to receive low-stock/expiration alerts and access supplier contact information so that I can reorder necessary pharmaceuticals before running out.

Acceptance Criteria:

Given a medication's quantity drops below the minimum defined threshold or is near expiration,

When the administrator views the pharmacy dashboard,

Then the system highlights the low-stock item with an alert badge and provides direct contact details for the associated supplier.

US_ADMIN_02: Manage Staff Accounts and Roles

User Story: As a Clinic Administrator, I want to create, edit, and deactivate staff accounts and assign roles so that each employee only accesses what their job requires.

Acceptance Criteria:

Given the administrator is on the user management page,

When they create an account with name, email, and role, or deactivate an existing account,

Then the system saves the change, the user's permissions match the assigned role from their next login, and deactivated users can no longer log in.

US_ADMIN_03: Define Veterinarian Schedules

User Story: As a Clinic Administrator, I want to define each veterinarian's working hours and time off so that only valid time slots are offered for appointments.

Acceptance Criteria:

Given the administrator is on the staff schedule page,
When they set a veterinarian's weekly hours and time off and save,
Then the system stores the schedule and offers appointment slots only within the veterinarian's available hours.

US_ADMIN_04: Register Stock Entries and Suppliers

User Story: As a Clinic Administrator, I want to register incoming stock with its lot and expiration date and keep supplier and item records so that the inventory is accurate and organized by batch.

Acceptance Criteria:

Given the administrator is on the inventory management page,
When they register a purchase receipt with items, quantities, lot numbers, expiration dates, and supplier,
Then the system adds the stock, displays it grouped by lot and expiration date, and links each item to its supplier and minimum threshold.

All Users

US_USER_01: Log In Securely

User Story: As a registered user, I want to log in with my credentials so that I can access the features allowed for my role.

Acceptance Criteria:

Given the user is on the login page,
When they enter valid credentials,
Then the system starts a session and shows the home screen for their role.
Given the user is on the login page,
When they enter invalid credentials,
Then the system shows an error message and does not start a session.
Given the user has an active session,
When they log out or remain inactive beyond the session limit,
Then the system invalidates the session and requires a new login.

US_USER_02: Recover My Password

User Story: As a registered user, I want to reset my password through my email so that I can regain access if I forget it.

Acceptance Criteria:

Given the user is on the login page,
When they click "Forgot Password" and enter their registered email,
Then the system sends a time-limited password reset link to that email.

3. User Story Mapping

The map groups the user stories by user activity (columns) and prioritizes them into three releases (rows).

VetCare: User Story Map					
Columns are user activities from left to right. Rows are releases in priority order. 20 user stories.					
	Access & Accounts	Clients & Patients	Appointments	Hospitalization	Pharmacy & Inventory
Release 1: Core clinic operations Accounts, client and patient records, appointment booking with room assignment. 9 stories	US_USER_01 Log in securely All users FR_02, FR_03, FR_05	US_RECEPTIONIST_01 Register owner and pets Receptionist FR_07, FR_08	US_ADMIN_03 Define veterinarian schedules Administrator FR_19		
	US_OWNER_03 Create portal account Pet Owner FR_01	US_RECEPTIONIST_03 Search owners and patients Receptionist FR_13	US_OWNER_01 Request appointment online Pet Owner FR_16		
	US_ADMIN_02 Manage staff and roles Administrator FR_06	US_VET_01 Species-specific medical record Veterinarian FR_09, FR_10, FR_12	US_RECEPTIONIST_02 Confirm and assign room Receptionist FR_17, FR_18		
Release 2: Hospitalization and pharmacy Inpatient care, vaccines, prescriptions and inventory control. 8 stories	US_USER_02 Recover password All users FR_04	US_VET_03 Record vaccinations Veterinarian FR_14		US_VET_05 Admit and discharge patients Veterinarian FR_22	US_ADMIN_04 Register stock and suppliers Administrator FR_27, FR_28, FR_31
				US_VET_02 Monitor map and rounds Veterinarian FR_21, FR_23, FR_24	US_VET_04 Issue prescription Veterinarian FR_15, FR_29
				US_ASSISTANT_01 Daily care and supplies Vet. Assistant FR_23, FR_29	US_ADMIN_01 Stock alerts and supplier lookup Administrator FR_30, FR_32
Release 3: Owner portal and reminders Owner self-service and automated notifications. 3 stories		US_OWNER_02 View simplified medical sheet Pet Owner FR_11	US_OWNER_04 Cancel or reschedule Pet Owner FR_20		
			US_OWNER_05 Receive reminders Pet Owner FR_25, FR_26		
Cross-cutting requirements: NFR_06 to NFR_10 apply to every release. NFR_01, NFR_02, NFR_08 (auditability) and NFR_10 (CI/CD) are needed from Release 1. NFR_03, NFR_04 and NFR_05 are verified in the release that delivers the related module.					

4. References

- Berné, M. (2023, March 8). *Criterios de aceptación: ejemplos para elaborarlos* [Acceptance criteria: examples for developing them]. Scrum Manager Blog. <https://www.scrummanager.com/blog/2023/03/criterios-de-aceptacion-definicion-y-ejemplos/>
- MacKay, J. (2019, January 14). *A guide to user story mapping: Templates and examples (How to map user stories)*. Planio. <https://plan.io/blog/user-story-mapping/>
- Rehkopf, M. (n.d.). *Historias de usuario con ejemplos y plantilla* [User stories with examples and template]. Atlassian. <https://www.atlassian.com/es/agile/project-management/user-stories>

5. Use of Artificial Intelligence

The team used an AI assistant (Claude, by Anthropic) as a support tool during the preparation of this workshop. Its use covered the following tasks:

- Drafting and improving the wording of requirements and user stories and translating the document into English.
- Reviewing the draft to identify missing requirements and inconsistencies between requirements and user stories and suggesting additions.

The project idea, the scope, the selection of the roles and modules, and the final decision on which requirements, user stories, and releases to include were made by the team members, who reviewed and edited all AI-generated content.